Elijah Akpan

eakpan@wpi.edu | Boston, MA | (508)-309-0096 | https://www.linkedin.com/in/elijah-akpan-9675a1235/ | portfolio

Education

Worcester Polytechnic Institute | Worcester, MA Exp. Graduation Date: 2028
B.S/M.S in Mechanical Engineering GPA: 3.7
Relevant Coursework: Heat Transfer, Thermodynamics, Differential Equations, Advanced Engineering Design, Dynamic Systems

Skills

Softwares: SolidWorks, MasterCAM, AutoCAD, LabView, CATIA, KiCAD, Vericut, Vigrade, Siemens NX
Programming Languages: MATLAB, Python, SCPI, G-code, C++
Hardware: Arduino, Haas CNC Machines, BambuLab, Raspberry Pi, Prusa, Fanuc controlled VTL CNC machines

Professional Experience

Mechanical Engineering Intern, **Polamer Precision** | New Britain, CT May 2026 - Aug 2026

- Developed and optimized manufacturing processes for engine components used in 30000+ pounds of thrust aerospace engines.
- Programmed complex CAM toolpaths using MasterCAM for engine components in both turning and milling.
- Reduced production errors by optimizing machine and CAM simulation through creating machine templates for Vericut. Reverse engineered VTL machines using SolidWorks and wrote g-code/C++ macros for tool changing and turret actuation.

Mechanical Engineering Intern, **TTM Technologies** | Syracuse, NY Aug 2025 - Dec 2025

- Designed and simulated performance of RF/PCB components for fighter jets.
- Helped design a new 3dB hybrid coupler using thermal FEA simulations to maximize both thermal and RF performance.
- Optimized quality process for RF parts by saving engineers 100+ hours of quality work by developing a MATLAB GUI that uses shape recognition to automatically detect defected vias from folders containing 200+ images.

Autonomous Vehicle Researcher, **Autonomous Vehicle Mobility Institute** | Worcester, MA Oct 2024 - Present

- Design and manufacture components for an autonomous vehicle laboratory that specializes in off-road vehicles.
- Led the design and manufacturing of a specialized vehicle simulation cave by modifying T-slots, sourcing plywood for screen, machining projector mount, and designing PLA brackets used for support. The lab secured a $65,000 dollar grant due to the cave.
- Helped upgrade a $100,000-dollar unmanned ground vehicle by designing and manufacturing the suspension system, mounting bracket for DC motors, and the LiFePO4 battery cage.

Teaching Assistant, **Worcester Polytechnic Institute, Washburn Labs** | Worcester, MA Feb 2025 - Present

- Lead 2 lab sections consisting of 30+ students on how to manufacture their own 10 piece Stirling Engine using MasterCAM & Haas Mini Mills/Lathes CNC machines.

Extracurriculars

Performance Lead, **WPI Formula SAE |** Worcester, MA Aug 2024 - Present

- Leading the performance simulation and vehicle dynamics of a 230 horsepower electric FSAE racecar
- Designed a high performance suspension system from ground up using OptimumKinematics, Vigrade, and SolidWorks.
- Led the development of a polycarbonate enclosure with a $1000 dollar budget that protects, cools, and simultaneously charges a total of 8 highly energized lithium ion segments that generate a combined 400 volts.

Projects

Engine Simulator For Toy Ride On Cars July 2025 – Present

- Invented a (PATENT PENDING) engine simulator for ride on cars that replicates engine noises of the real car based on pedal tracking. Wrote the control algorithm and audio processing effects using python and designed the custom PCB using KiCAD.
- Raised $2000+ dollars so far for the invention.

Torque Vectoring/VI-grade Simulation | FSAE TEAM May 2026 - Present

- Designing torque vectoring controller based on Simulink control algorithm using yaw rate calculations from CAN data.
- Built ViCarRealTime simulation using python to model racecar, track layout, and loads to simulate performance.